

Functional Test Cases Report

Pharmacy Management System - Backend API functional test cases covering authentication, admin management, pharmacy orders, warehouse operations, sales, expenses, feedback, ratings, and statistics.

9.5.1 Authentication and Account Security

This section documents the functional test cases related to authentication, OTP verification, profile access, and password change for the Pharmacy Management System. These tests verify that users can securely access the system according to their roles.

Test Case 1: Admin Login

Preconditions: Admin account exists

Test Steps: 1. Enter valid email/phone and password; 2. Send login request

Expected Results: Admin is authenticated and token is returned

Actual Results: Admin logged in successfully

Status: Pass

Test Case 2: Pharmacy Login Sends OTP

Preconditions: Pharmacy account exists

Test Steps: 1. Enter valid login and password; 2. Submit login request

Expected Results: OTP is sent to pharmacy email and no token is created yet

Actual Results: OTP sent successfully

Status: Pass

Test Case 3: Warehouse Login Sends OTP

Preconditions: Warehouse account exists

Test Steps: 1. Enter valid login and password; 2. Submit login request

Expected Results: OTP is sent to warehouse email and no token is created yet

Actual Results: OTP sent successfully

Status: Pass

Test Case 4: Verify Pharmacy OTP

Preconditions: OTP was sent to pharmacy email

Test Steps: 1. Enter valid OTP; 2. Submit verify request

Expected Results: Pharmacy token is generated

Actual Results: Token returned successfully

Status: Pass

Test Case 5: Verify Warehouse OTP

Preconditions: OTP was sent to warehouse email

Test Steps: 1. Enter valid OTP; 2. Submit verify request

Expected Results: Warehouse token is generated

Actual Results: Token returned successfully

Status: Pass

Test Case 6: Invalid OTP Verification

Preconditions: OTP was sent

Test Steps: 1. Enter wrong OTP; 2. Submit verify request

Expected Results: Error message is returned and token is not created

Actual Results: Error displayed correctly

Status: Pass

Test Case 7: Change Password

Preconditions: User is authenticated

Test Steps: 1. Enter old password; 2. Enter new password; 3. Confirm new password

Expected Results: Password is changed successfully

Actual Results: Password updated successfully

Status: Pass

9.5.2 Admin Management

This section verifies the admin functionalities for managing pharmacies, warehouses, admins, and system reports.

Test Case 8: Create Pharmacy

Preconditions: Admin is authenticated

Test Steps: 1. Send pharmacy data; 2. Submit create request

Expected Results: Pharmacy account is created

Actual Results: Pharmacy created successfully

Status: Pass

Test Case 9: Create Warehouse

Preconditions: Admin is authenticated

Test Steps: 1. Send warehouse data; 2. Submit create request

Expected Results: Warehouse account is created

Actual Results: Warehouse created successfully

Status: Pass

Test Case 10: View Pharmacies

Preconditions: Admin is authenticated

Test Steps: 1. Send GET pharmacies request

Expected Results: List of pharmacies is returned

Actual Results: Pharmacies retrieved successfully

Status: Pass

Test Case 11: View Warehouses

Preconditions: Admin is authenticated

Test Steps: 1. Send GET warehouses request

Expected Results: List of warehouses is returned

Actual Results: Warehouses retrieved successfully

Status: Pass

Test Case 12: View System Reports

Preconditions: Admin is authenticated

Test Steps: 1. Request counts and regional reports

Expected Results: Admin receives system statistics

Actual Results: Reports displayed successfully

Status: Pass

9.5.3 Pharmacy Inventory and Orders

This section covers pharmacy inventory management, ordering from warehouses, and the order assistant feature.

Test Case 13: Add Product to Pharmacy Inventory

Preconditions: Pharmacy is authenticated

Test Steps: 1. Enter product barcode and stock data; 2. Submit request

Expected Results: Product is added to pharmacy inventory

Actual Results: Product saved successfully

Status: Pass

Test Case 14: View Pharmacy Inventory

Preconditions: Pharmacy has products

Test Steps: 1. Send GET inventory request

Expected Results: Pharmacy products are returned

Actual Results: Inventory displayed successfully

Status: Pass

Test Case 15: Add Item to Order Cart

Preconditions: Product exists in selected warehouse

Test Steps: 1. Select warehouse; 2. Add product and quantity

Expected Results: Item is added to order cart

Actual Results: Cart updated successfully

Status: Pass

Test Case 16: Update Order Cart Quantity

Preconditions: Item exists in order cart

Test Steps: 1. Change item quantity; 2. Submit update request

Expected Results: Quantity is updated if stock is available

Actual Results: Quantity updated successfully

Status: Pass

Test Case 17: Checkout Order Cart

Preconditions: Order cart contains valid items

Test Steps: 1. Submit checkout request

Expected Results: Order is created with pending status and stock is reserved

Actual Results: Pending order created successfully

Status: Pass

Test Case 18: Generate Order Assistant Proposal

Preconditions: Pharmacy has low/out-of-stock products

Test Steps: 1. Request assistant proposal

Expected Results: System suggests medicines and selects best warehouse

Actual Results: Proposal generated successfully

Status: Pass

Test Case 19: Apply Order Assistant Proposal

Preconditions: Assistant proposal exists

Test Steps: 1. Submit suggested warehouse and items; 2. Apply proposal

Expected Results: Suggested items are added to order cart

Actual Results: Proposal applied successfully

Status: Pass

9.5.4 Warehouse Operations

This section verifies warehouse inventory and order processing features.

Test Case 20: Add Product to Warehouse Inventory

Preconditions: Warehouse is authenticated

Test Steps: 1. Enter product and stock data; 2. Submit request

Expected Results: Product is added or updated in warehouse inventory

Actual Results: Product saved successfully

Status: Pass

Test Case 21: View Warehouse Orders

Preconditions: Warehouse is authenticated

Test Steps: 1. Send GET warehouse orders request

Expected Results: All pharmacy orders for the warehouse are returned

Actual Results: Orders retrieved successfully

Status: Pass

Test Case 22: Approve Pharmacy Order

Preconditions: Pending order exists

Test Steps: 1. Select pending order; 2. Approve order

Expected Results: Order status becomes approved

Actual Results: Order approved successfully

Status: Pass

Test Case 23: Reject Pharmacy Order

Preconditions: Pending order exists

Test Steps: 1. Select pending order; 2. Reject order

Expected Results: Order status becomes rejected and reserved stock is released

Actual Results: Order rejected successfully

Status: Pass

9.5.5 Order Receiving and Sales

This section covers receiving approved orders and pharmacy sales checkout behavior.

Test Case 24: Receive Approved Order

Preconditions: Order status is approved

Test Steps: 1. Pharmacy selects approved order; 2. Submit receive request

Expected Results: Warehouse stock decreases and pharmacy stock increases

Actual Results: Order received successfully

Status: Pass

Test Case 25: Add Item to Sales Cart

Preconditions: Product exists in pharmacy inventory

Test Steps: 1. Scan product barcode; 2. Add item to sales cart

Expected Results: Item is added if stock is available

Actual Results: Sales cart updated successfully

Status: Pass

Test Case 26: Checkout Sales Cart

Preconditions: Sales cart contains valid items

Test Steps: 1. Enter paid total; 2. Submit checkout

Expected Results: Sales invoice is created and pharmacy stock decreases

Actual Results: Sale recorded successfully

Status: Pass

Test Case 27: Confirm High Discount Sale

Preconditions: Discount is 20% or more

Test Steps: 1. Submit checkout; 2. Confirm discount

Expected Results: Sale is recorded after confirmation

Actual Results: High discount sale confirmed successfully

Status: Pass

9.5.6 Expenses, Feedback, Ratings, and Statistics

This section verifies supporting business features such as expenses, feedback, warehouse ratings, and statistics.

Test Case 28: Create Expense Invoice

Preconditions: Pharmacy or warehouse is authenticated

Test Steps: 1. Enter amount, creator name, and description; 2. Submit request

Expected Results: Expense invoice is created

Actual Results: Expense invoice created successfully

Status: Pass

Test Case 29: Rate Warehouse

Preconditions: Pharmacy has received an order from warehouse

Test Steps: 1. Select warehouse; 2. Submit rating from 1 to 5

Expected Results: Rating is saved or updated and average rating changes

Actual Results: Warehouse rated successfully

Status: Pass

Test Case 30: View Statistics Summary

Preconditions: User is authenticated

Test Steps: 1. Request statistics summary

Expected Results: Revenue, expenses, and net summary are returned

Actual Results: Statistics displayed successfully

Status: Pass